

Derivación Semántica Formal – Notación Nielson con Estados σ_k

Rama 1 de 1 (NS)

$$\mathcal{T}_1 = \langle x := 1, \sigma_0 \rangle \longrightarrow \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] \text{ [ass}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_2 = \frac{\langle \varepsilon, \sigma_2 \rangle \longrightarrow_D \sigma_2 \text{ [none}_{\text{ns}}]}}{\langle \mathbf{var} \ y := 10 ; , \sigma_3 \rangle \longrightarrow_D \sigma_3[y \mapsto \mathcal{A}[\![10]\!]\sigma_3] \text{ [var}_{\text{ns}}]}}$$

$\Downarrow$

$$\mathcal{T}_3 = \frac{\mathcal{T}_2}{\langle \mathbf{var} \ x := 5 ; \mathbf{var} \ y := 10 ; , \sigma_1 \rangle \longrightarrow_D \sigma_1[x \mapsto \mathcal{A}[\![5]\!]\sigma_1] \text{ [var}_{\text{ns}}]}}$$

$\Downarrow$

$$\mathcal{T}_4 = \langle x := 99, \sigma_2 \rangle \longrightarrow \sigma_2[x \mapsto \mathcal{A}[\![99]\!]\sigma_2] \text{ [ass}_{\text{ns}}]$$

$\Downarrow$

$$\mathcal{T}_5 = \langle z := 11, \sigma_5 \rangle \longrightarrow \sigma_5[z \mapsto \mathcal{A}[\![11]\!]\sigma_5] \text{ [ass}_{\text{ns}}]$$

$\Downarrow$

$$\begin{array}{c}
\mathcal{T}_6 = \frac{\mathcal{T}_4 \quad \mathcal{T}_5}{\langle (x := 99 ; z := 11), \sigma_2 \rangle \longrightarrow \sigma_7} [\text{comp}_{\text{ns}}] \\
\Downarrow \\
\mathcal{T}_7 = \frac{\mathcal{T}_3 \quad \mathcal{T}_6}{\langle S_1, \sigma_1 \rangle \longrightarrow \sigma_7 [\text{DV}(\text{var } x := 5 ; \text{var } y := 10 ;) \mapsto \sigma_1]} [\text{block}_{\text{ns}}] \\
\Downarrow \\
\mathcal{T}_8 = \frac{\mathcal{T}_1 \quad \mathcal{T}_7}{\langle S_2, \sigma_0 \rangle \longrightarrow \sigma_8} [\text{comp}_{\text{ns}}]
\end{array}$$

Sentencias Abreviadas

$$\begin{aligned}
S_1 &= \mathbf{begin} \, (\text{var } x := 5 ; \text{var } y := 10 ;) \, (x := 99 ; z := 11) \, \mathbf{end} \\
S_2 &= (x := 1 ; \mathbf{begin} \, (\text{var } x := 5 ; \text{var } y := 10 ;) \, (x := 99 ; z := 11) \, \mathbf{end})
\end{aligned}$$

Leyenda de Estados y Sustituciones

$$\begin{aligned}
\sigma_0 &\equiv \emptyset \quad (\text{Estado inicial vacío}) \quad (\sigma_{\text{ini}} - \mathbf{Estado Inicial}) \\
\sigma_1 &\equiv \sigma_0[x \mapsto \mathcal{A}[[1]]\sigma_0] = \{x \mapsto 1\} \\
\sigma_2 &\equiv \sigma_1[x \mapsto \mathcal{A}[[5]]\sigma_1] = \{x \mapsto 5, y \mapsto 10\} \\
\sigma_3 &\equiv \{x \mapsto 5\} \\
\sigma_4 &\equiv \sigma_3[y \mapsto \mathcal{A}[[10]]\sigma_3] = \{x \mapsto 5, y \mapsto 10\} \\
\sigma_5 &\equiv \sigma_2[x \mapsto \mathcal{A}[[99]]\sigma_2] = \{x \mapsto 99, y \mapsto 10\} \\
\sigma_6 &\equiv \sigma_5[z \mapsto \mathcal{A}[[11]]\sigma_5] = \{x \mapsto 99, y \mapsto 10, z \mapsto 11\} \\
\sigma_7 &\equiv \{x \mapsto 99, y \mapsto 10, z \mapsto 11\} \\
\sigma_8 &\equiv \sigma_7[\text{DV}(\text{var } x := 5 ; \text{var } y := 10 ;) \mapsto \sigma_1] = \{x \mapsto 1, z \mapsto 11\} \quad (\sigma_{\text{fin}} - \mathbf{Estado Final})
\end{aligned}$$